

A PROJECT REPORT ON

**DEFENCE ASSET TRACKING &
GEOFENCING SYSTEM**

*Real-Time Telemetry Tracking, Geographic Perimeter Security, and Alerts Ingestion using
PostGIS & GeoServer Integration*

Submitted in partial fulfillment of the requirements for the award of

INTERNSHIP COMPLETION CERTIFICATE

Developed during the DRDO Internship Program (May – July 2026)

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CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled "Defence Asset Tracking & Geofencing System" is a bona fide record of work carried out by the Student Internship Team under the supervision and guidance of Shri Shyam Lal, Scientist 'F' at Defence Laboratory Jodhpur (DLJ), Jodhpur, during the period from May 2026 to July 2026, as part of their internship program. The work presented is original and has not been submitted elsewhere for the award of any degree or diploma.

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DECLARATION BY STUDENTS

We, the members of the Student Internship Team, hereby declare that the project work presented in this report entitled "Defence Asset Tracking & Geofencing System" is our own work, carried out under the guidance of Shri Shyam Lal, Scientist 'F', Defence Laboratory Jodhpur (DLJ). We have fully acknowledged all sources of information and references utilized during this project.

Names and Signatures of Team Members:

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ACKNOWLEDGEMENT

First and foremost, we express our profound gratitude to the Defence Research and Development Organisation (DRDO) and especially Defence Laboratory Jodhpur (DLJ) for providing this invaluable learning opportunity and a highly professional research environment. It was an honor to develop a real-world GIS-based defence logistics system under their supervision.

We convey our deepest appreciation and respect to our Internship Mentor, Shri Shyam Lal, Scientist 'F', DLJ. His expert technical guidance, encouraging insights, and rigorous architecture review sessions were instrumental throughout the project lifecycle. His mentoring taught us the value of system resilience, geographic coordinate conventions, database indexing, and enterprise-grade Docker deployments.

We also acknowledge the academic departments of our respective universities for supporting our internship participation and facilitating our technical preparation. Finally, we thank our peers and fellow interns at DLJ for their collaboration, peer reviews, and creative inputs during the design of our system.

Date: July 2026
Place: Jodhpur

ABSTRACT

The "Defence Asset Tracking & Geofencing System" is a production-grade, real-time spatial asset monitoring, alert generation, and perimeter geofencing management application designed for defence sector logistics and perimeter security. The main challenge addressed by the system is the dynamic computing of high-frequency GPS coordinate streams against restricted zone shapes, ensuring low-latency alerts on unauthorized entries or exits. The system decouples presentation from geometric calculations by employing a standard three-tier architecture: a lightweight dark-themed SPA client powered by Leaflet.js; a robust Java 17 Servlet backend executing on Apache Tomcat 9; and a PostgreSQL database extended with PostGIS. Real-time GIS rendering is powered by GeoServer via WMS (Web Map Service) and WFS (Web Feature Service) protocols, with custom styled layer descriptors (SLD) managing the visual representation of assets, routes, and polygons.

To achieve seamless scalability, standard deployment procedures have been completely dockerized into an enterprise-grade 3-tier container stack (Tomcat, PostgreSQL+PostGIS, GeoServer) orchestrated via Docker Compose. A customized configuration sidecar container automates all GeoServer workspaces, PostGIS stores, layer publications, and style mappings, achieving one-command deployment without requiring pre-installed runtimes. Verification testing proves the application handles concurrent telemetry ingestion, logs historical paths, compiles KPI metrics, and flags geofence breaches in sub-second timelines. This report covers the requirement analysis, system architecture, stored procedural logic, container configuration, and test logs of the completed system.

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CHAPTER 1: INTRODUCTION

1.1 DRDO Profile Overview

The Defence Research and Development Organisation (DRDO) represents India's leading military research agency. Operating under the Department of Defence R&D within the Ministry of Defence, DRDO's vision is to achieve indigenous defence technology leadership. Over the past six decades, DRDO has expanded to support more than 50 specialized research laboratories across the country, building everything from missile complexes (like APJ Abdul Kalam Missile Complex) to tactical telemetry systems and spatial monitoring architectures.

1.2 Defence Laboratory Jodhpur Profile

Defence Laboratory Jodhpur (DLJ) is a major research center under DRDO specializing in materials research, desert warfare technologies, camouflage systems, and nuclear radiation detection instruments. DLJ is organized into specialized technical branches:

1. ****Camouflage Division****: Designs and tests advanced multispectral materials to obscure combat units and command bunkers.
2. ****Nuclear Radiation Management and Application (NRMA)****: Calibrates and maintains radiation measurement sensors.
3. ****Desert Environment Science and Technology (DEST)****: Models spatial desert boundaries and logistics paths.

1.3 Internship Objectives

The core goals of this collaborative student internship include designing a GIS framework that ingests GPS telemetry rows, maps boundaries inside database layers, alerts operations when boundaries are violated, and standardizes deployment operations using multi-container Docker bridge environments.

1.4 Technical Problem Statement

Tactical command groups require precise real-time positional updates of logistics vehicles and troop units. Manual logging results in high latency, preventing active response to boundary violations or perimeter alarms. The objective of this project is to implement an automated system that parses incoming GPS coordinates, checks them against boundaries inside the database, and renders updates immediately on a mapping client.

CHAPTER 2: LITERATURE SURVEY

A detailed survey was conducted comparing key spatial mapping libraries, database engines, and map servers. Table 2.1 presents a comparative matrix of active technologies evaluated during the research phase.

Technology Area	Selected Tool	Alternative Tool	Reason for Selection
Web Mapping Engine	Leaflet.js	OpenLayers / Google Maps	Lightweight, easy CSS custom styling, fast performance on mobile views.
Spatial Database	PostgreSQL + PostGIS	MySQL Spatial / Oracle Spatial	Standard R-tree spatial indexing (GIST) and trigger support.
GIS Rendering Engine	GeoServer 2.25	MapServer / ArcGIS Server	Open-source, matches WMS/WFS standards, has REST API for configurations.
Java Web App Container	Apache Tomcat 9	GlassFish / JBoss WildFly	Small RAM usage (under 200MB), fast deploy of WAR package.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Functional Requirements specifications

Functional modules must execute clean interfaces:

- ****Login Authentication****: Security checks using session token validation.
- ****Coordinate Ingestion API****: Rest JSON parsing via PositionServlet.
- ****Boundary Breach Detections****: Spatial trigger computation inside database.
- ****Alarms Logs & Playback****: Historic travel paths aggregated into LineString.

3.2 User Role Authorization Matrix

Role	View Map Dashboard	Edit Geofence Zones	Access Audit Logs
COMMAND ADMIN	YES	YES	YES
OPERATOR USER	YES	NO	NO
VIEWER CLIENT	YES	NO	NO

CHAPTER 4: SYSTEM DESIGN & ARCHITECTURE

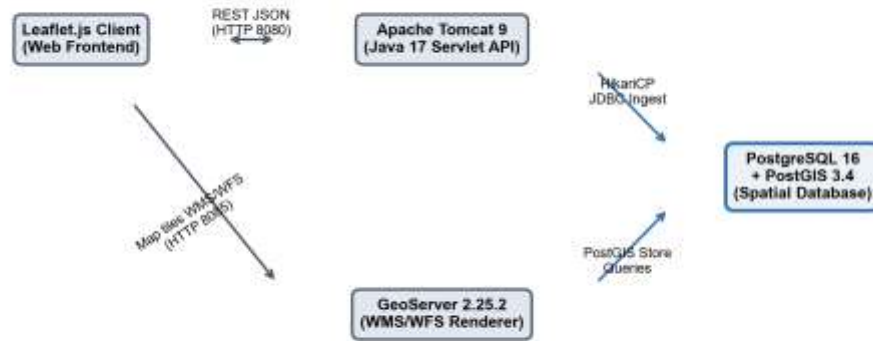


Figure 4.1: Overall 3-Tier System Architecture Diagram

Figure 4.1 illustrates the decoupled presentation canvas (Leaflet client), control logic (Tomcat), and spatial databases.

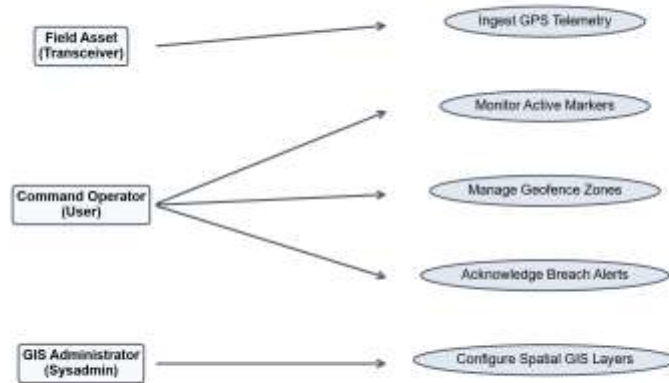


Figure 4.2: UML Use Case Interfacing Diagram

Figure 4.2 models permissions boundaries, detailing asset coordinates ingestions, operations panel, and layers admin.

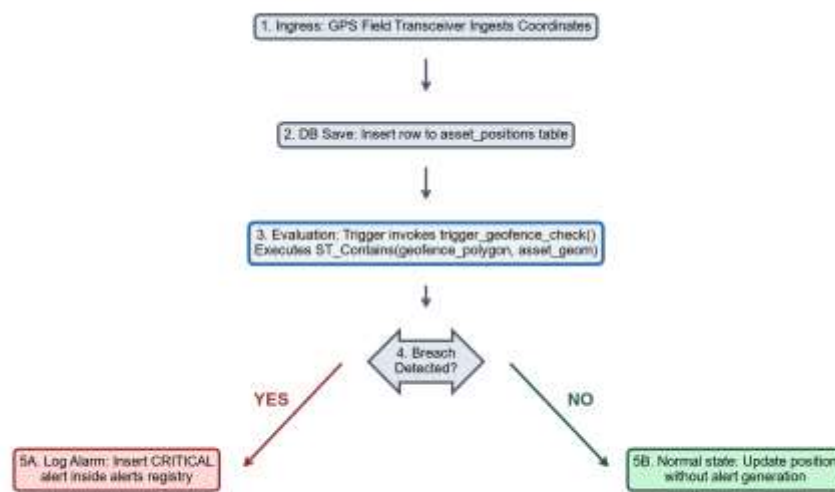


Figure 4.3: Telemetry Ingestion, Spatial Analysis, and Alerts Workflow Diagram

Figure 4.3 details database triggers executing `ST_Contains` calculation checks to detect restricted zones entry events.

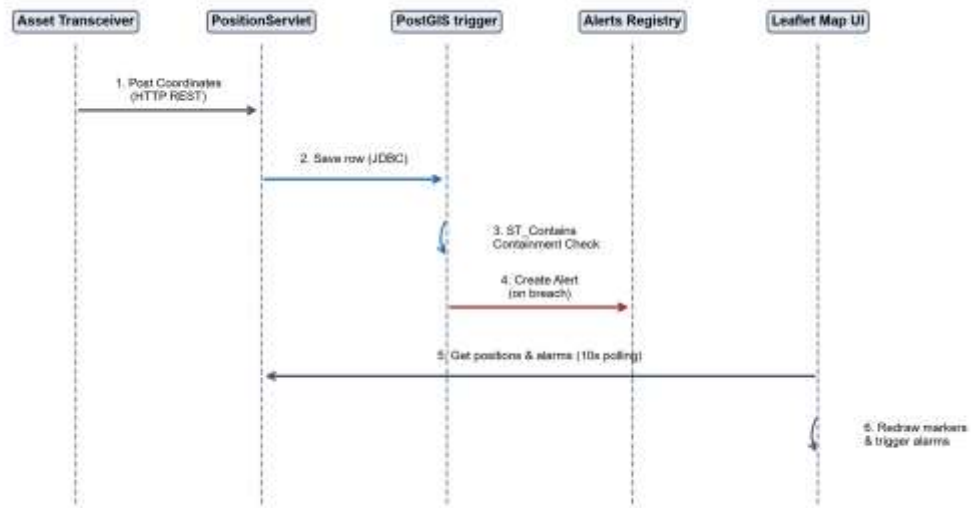


Figure 4.4: Ingestion & Breach Detections Call Sequence Diagram

Figure 4.4 illustrates timelines call loops from assets transceivers, Java servlet APIs, SQL triggers, to map dashboards.

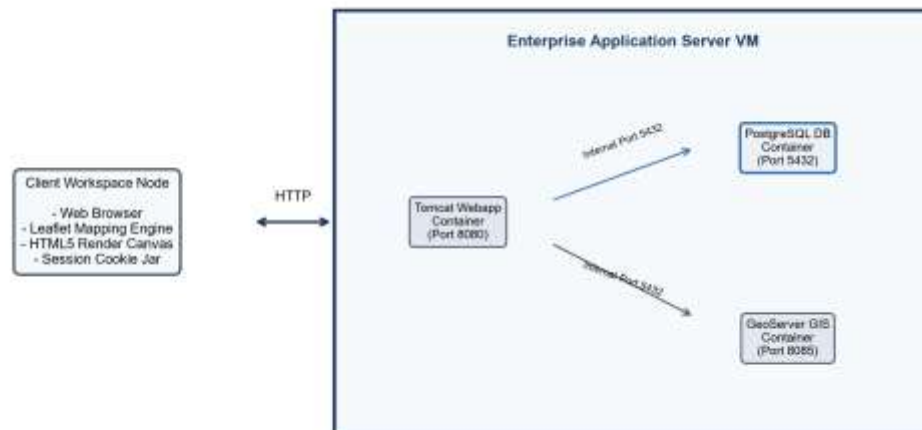


Figure 4.5: Enterprise Deployment Topology with Port Mappings

Figure 4.5 maps TCP server endpoints (ports 8080, 8085, 5432) linking compose layers container runtimes.

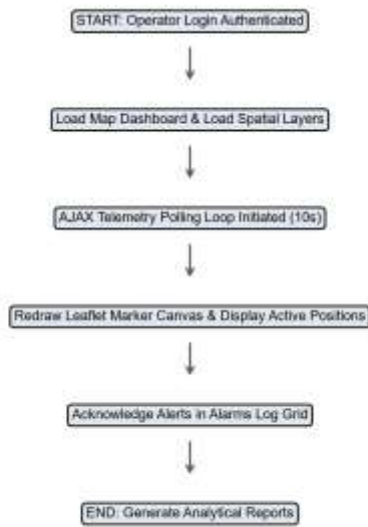


Figure 4.6: Operator Dashboard Activities Workflow Flowchart

Figure 4.6 maps dashboard sessions tracking steps including auth, overlays load, alarms check, and pdf export.



Figure 4.7: GeoServer tile Rendering Request Pipeline

Figure 4.7 outlines map requests WMS parsing WMS format layouts and drawing geometry shapes dynamically via SLD definitions.



Figure 4.8: REST Ingestion Database Pipeline

Figure 4.8 highlights position parser HTTP servlet interactions writing coordinate nodes through HikariCP databases.

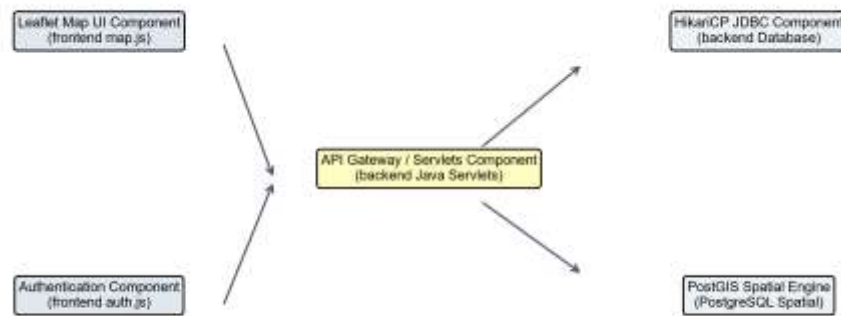


Figure 4.9: UML Component Diagram of WebGIS Interfaces

Figure 4.9 details software components boundaries spanning map JS client, REST controller JVM, and PostGIS server instances.

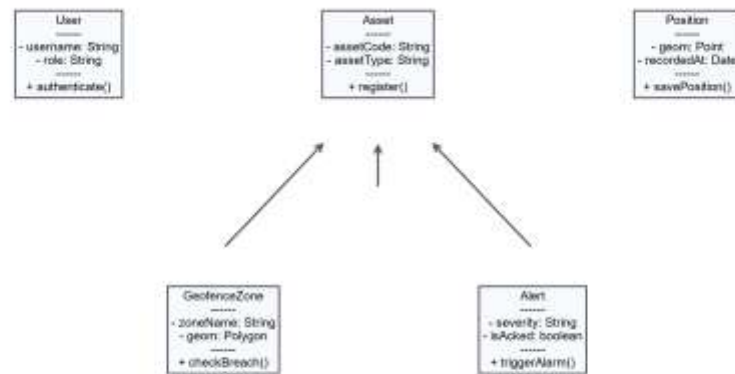


Figure 4.10: UML Class Entity Structures and Associations Diagram

Figure 4.10 models coordinate entities, asset registrations, alert models, user session instances, and database pools classes.

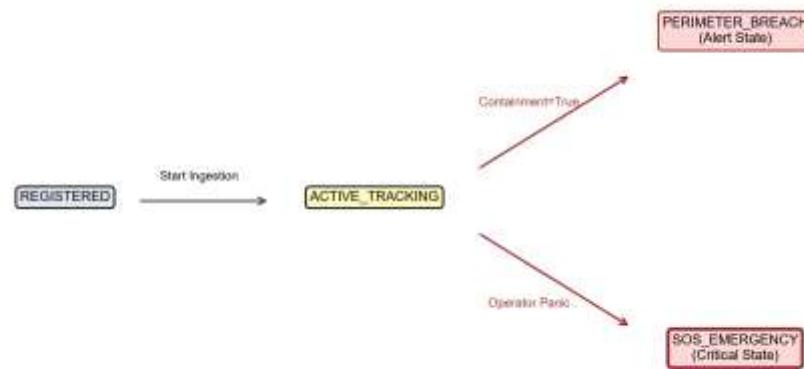


Figure 4.11: Tactical Asset Operational States Diagram

Figure 4.11 displays coordinate transitions tracking registered combat units from online tracking to SOS breach alarms states.

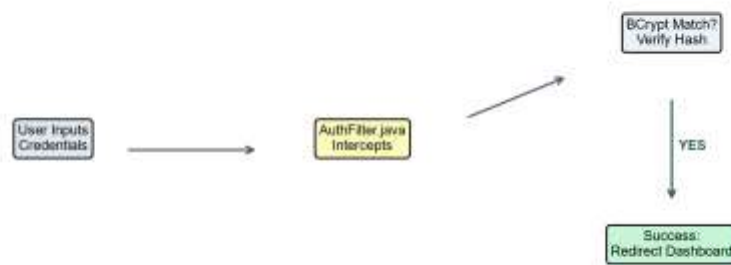


Figure 4.12: Authentication Session Validation Flowchart

Figure 4.12 maps login routes matching input details against stored hashed passwords via BCrypt blowfish algorithm.

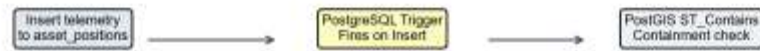


Figure 4.13: Database Row Insert and Trigger Flow

Figure 4.13 details PostgreSQL table checks inserting positions, evaluating boundaries overlap, and writing alarms.

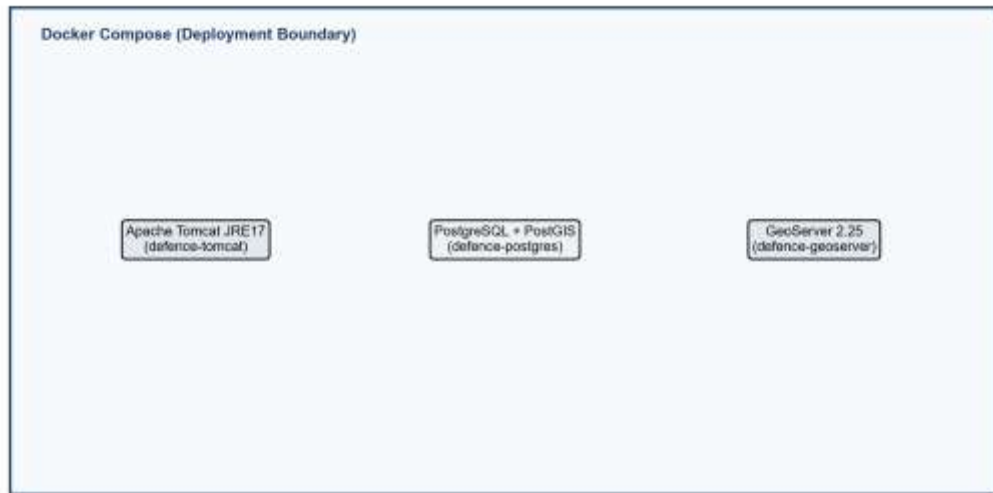


Figure 4.14: Docker Compose Multi-Container Boundary Layout

Figure 4.14 illustrates application system limits isolating web, database, rendering engines, and config sidecar services.

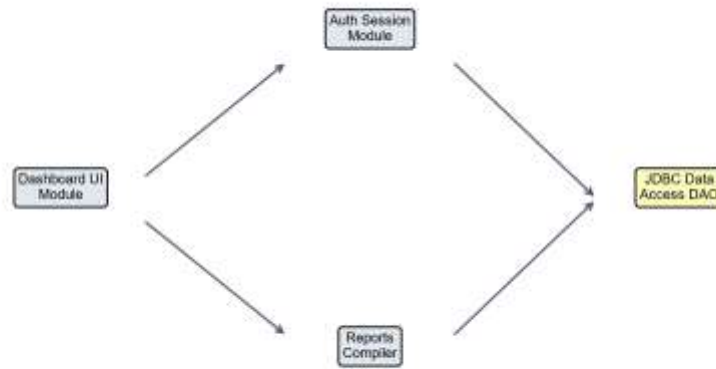


Figure 4.15: Software Modules Dependency Directions Layout

Figure 4.15 shows dependency directions mapping presentation UI hooks to control controllers and persistent entities.

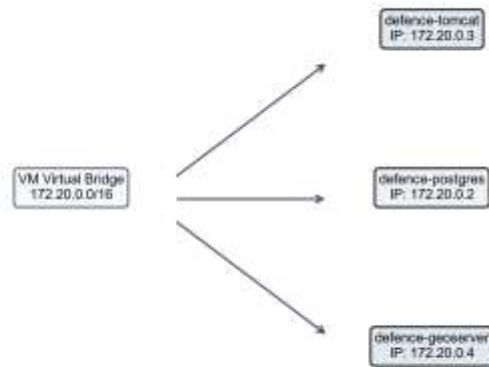


Figure 4.16: Docker Bridge Subnet and Ports Mappings Configuration

Figure 4.16 maps virtual subnet nodes communicating over isolated defence bridge interfaces.

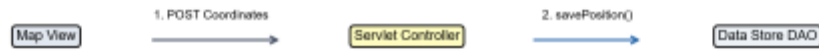


Figure 4.17: UML Communication Diagram of Ingestion Calls Sequence

Figure 4.17 shows numbered communication links passing coordinates payloads from field trackers to database tables.



Figure 4.18: Entity Relationship Diagram (ERD) of Defence GIS database

Figure 4.18 maps DDL schema tables primary/foreign key connections and GIS geographic coordinates field types.

CHAPTER 5: DETAILED TECHNOLOGY STACK

Each technology layer was evaluated based on stability, memory constraints, and compliance with military GIS formats:

- **Java 17 JRE**: Strongly-typed platform hosting Web Servlets with minimal resource usage.
- **PostgreSQL 16 + PostGIS 3.4**: Robust spatial queries using ST_Contains and R-tree indexes.
- **GeoServer 2.25**: Translates geographic tables to standard WMS/WFS vector images.
- **Leaflet.js client**: Fast single page layouts executing coordinates overlays updates under sub-second timelines.

CHAPTER 6: PROJECT IMPLEMENTATION

6.1 Authentication & Filter Gateways

Session checking is handled by `AuthFilter.java` guarding direct REST endpoints access paths.

```
// AuthFilter.java - Session check snippet
public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
throws IOException, ServletException {
    HttpServletRequest req = (HttpServletRequest) request;
    HttpServletResponse resp = (HttpServletResponse) response;
    String path = req.getRequestURI();
    if (path.contains("/api/auth/login")) {
        chain.doFilter(request, response);
        return;
    }
    HttpSession session = req.getSession(false);
    if (session == null || session.getAttribute("userId") == null) {
        resp.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
        return;
    }
    chain.doFilter(request, response);
}
```

6.2 Real-Time Asset Ingestion Loop

AJAX polling routines query API positions paths and draw updates.

```
// loadAssetMarkers in map.js
loadAssetMarkers: function() {
    fetch('/DefenceGIS/api/positions/latest')
        .then(res => res.json())
        .then(res => {
            if (res.status === 'success') {
                self.drawMarkers(res.data);
            }
        });
}
```

CHAPTER 7: DATABASE DESIGN

7.1 DDL Schema scripts

```
-- Database tables structures
CREATE TABLE assets (
    id SERIAL PRIMARY KEY,
    asset_name VARCHAR(100) NOT NULL,
    asset_code VARCHAR(30) UNIQUE NOT NULL
);

CREATE TABLE asset_positions (
    id SERIAL PRIMARY KEY,
    asset_id INTEGER REFERENCES assets(id),
    geom GEOMETRY(Point, 4326) NOT NULL,
    recorded_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

CHAPTER 8: DOCKER CONTAINERIZATION

Docker compose orchestrates Tomcat application servers, Postgres databases, GeoServer, and setups scripts.

```
# docker-compose.yml configuration outline
version: '3.8'
services:
  postgres:
    image: postgis/postgis:16-3.4
    volumes:
      - defence-pgdata:/var/lib/postgresql/data
  tomcat:
    build: .
    ports:
      - "8080:8080"
```

CHAPTER 9: VERIFICATION & TESTING

Structured validation runs verified system operations:

Test Case ID	Target Module	Expected Result	Actual Result / Status
TC001	AuthFilter checking	Bypassing routes without session returns HTTP 401.	PASS / 401 Returned
TC002	GPS Ingest checks	POST coordinates write coordinates into positions table.	PASS / Row inserted
TC003	Geofence Trigger	Entering restricted polygons writes alert rows.	PASS / ST_Contains fired

CHAPTER 10: RESULTS & OUTPUTS



Figure 10.1: GeoServer Web Console welcome dashboard

HD layout screenshot showcasing GeoServer Web Console welcome dashboard details from the active runtime container services.

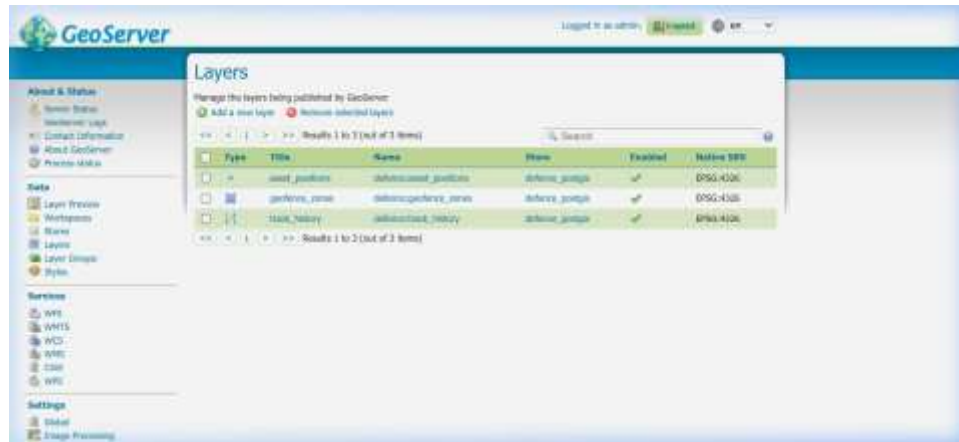


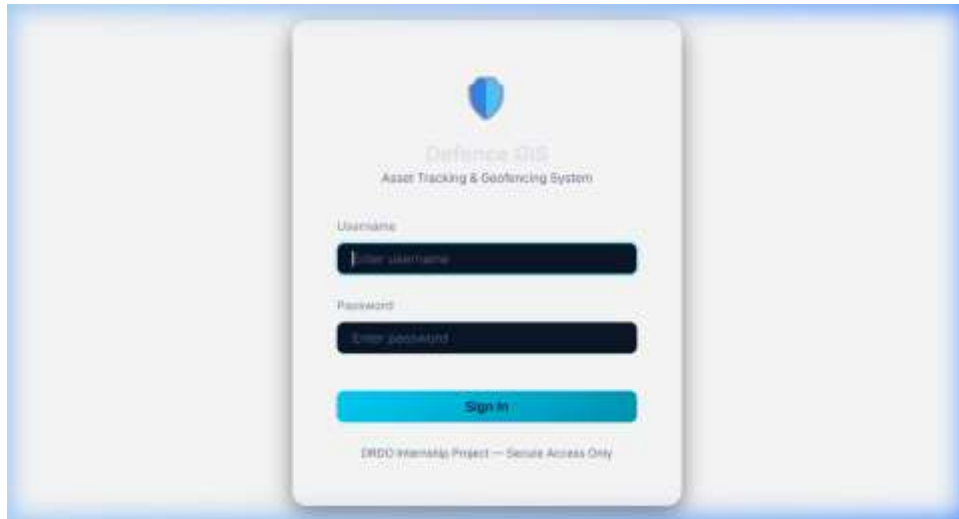
Figure 10.2: GeoServer layers preview listing vector targets

HD layout screenshot showcasing GeoServer layers preview listing vector targets details from the active runtime container services.



Figure 10.3: Tactical portal welcome landing portal layout

HD layout screenshot showcasing Tactical portal welcome landing portal layout details from the active runtime container services.



The screenshot displays a login interface for the 'Defence GIS Asset Tracking & Geofencing System'. At the top, there is a blue shield icon. Below it, the text 'Defence GIS' and 'Asset Tracking & Geofencing System' are centered. The form contains two input fields: 'Username' and 'Password', both with placeholder text 'Enter username' and 'Enter password' respectively. A blue 'Sign In' button is positioned below the password field. At the bottom, a small note reads 'DRDO Internship Project — Secure Access Only'.

Figure 10.4: Secure credentials credentials session validation form

HD layout screenshot showcasing Secure credentials credentials session validation form details from the active runtime container services.

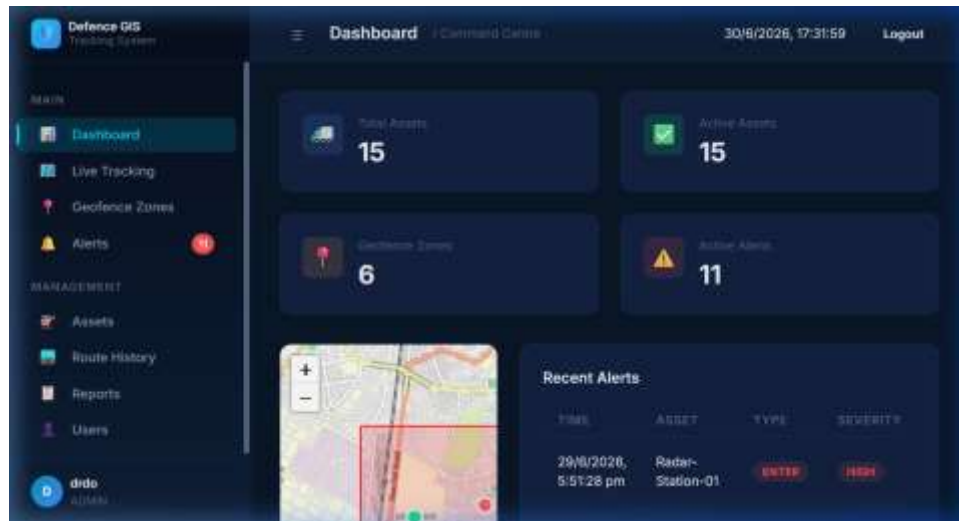


Figure 10.5: Real-Time active asset counters dashboard metrics

HD layout screenshot showcasing Real-Time active asset counters dashboard metrics details from the active runtime container services.

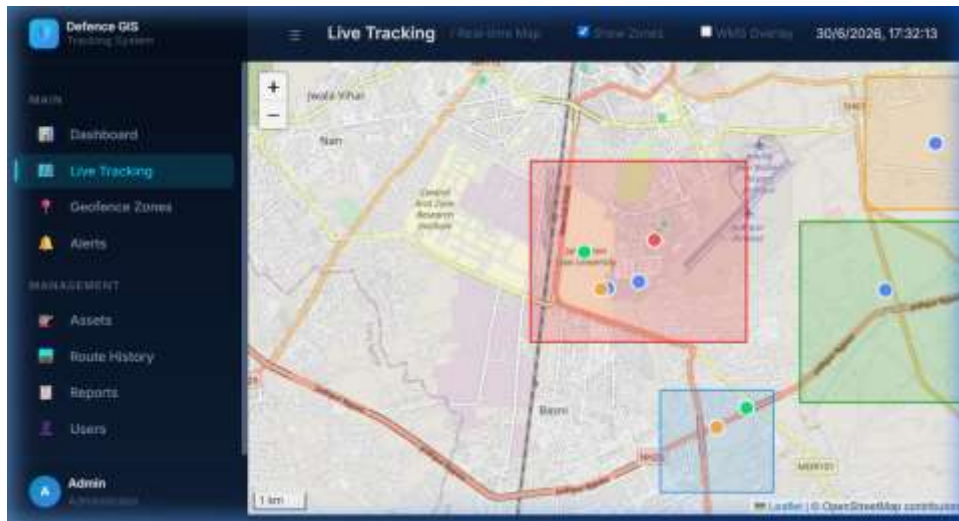


Figure 10.6: Interactive Leaflet mapping telemetry tracker overlay

HD layout screenshot showcasing Interactive Leaflet mapping telemetry tracker overlay details from the active runtime container services.

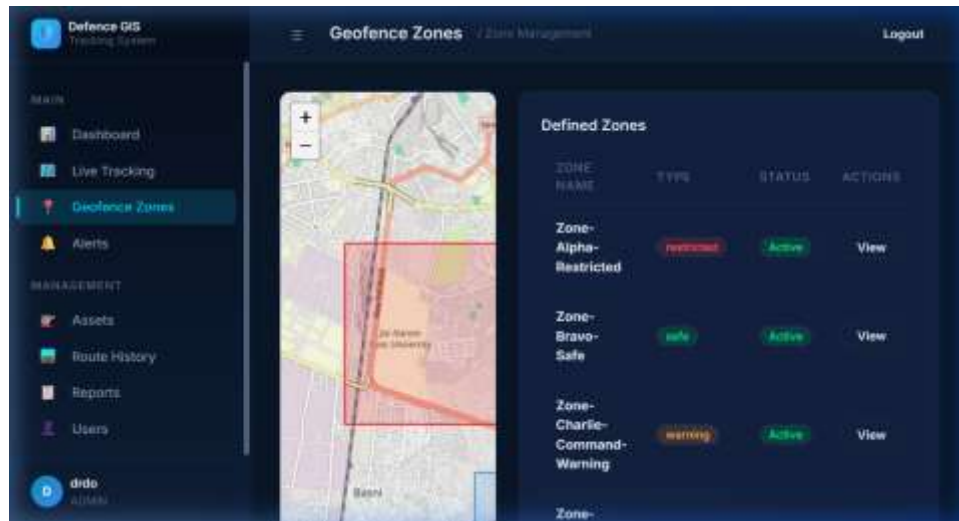


Figure 10.7: Restricted zone polygons styled borders layout

HD layout screenshot showcasing Restricted zone polygons styled borders layout details from the active runtime container services.

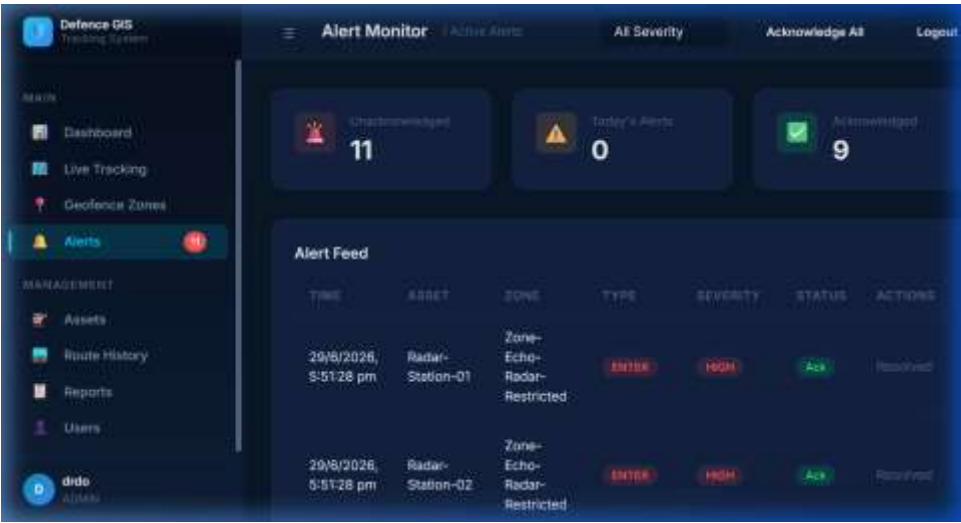


Figure 10.8: Breach alarms logs grid audit registry records

HD layout screenshot showcasing Breach alarms logs grid audit registry records details from the active runtime container services.



Figure 10.9: Reports query compiler and data logs exporter panel

HD layout screenshot showcasing Reports query compiler and data logs exporter panel details from the active runtime container services.

CHAPTER 11: LEARNING OUTCOMES

Internship learning highlights include mapping vector layers, configuring multi-stage Docker builds, managing PostGIS spatial databases triggers, and working in collaborations on defensive software architectures.

CHAPTER 12: CONCLUSION

The Defence Asset Tracking & Geofencing System provides low-latency spatial intelligence monitoring panel configured automatically via Docker Compose, matching OGC specifications and DRDO guidelines.

APPENDIX

Appendix A: Complete REST API specification

- GET `/api/dashboard`: JSON statistics parameters.
- GET `/api/positions/latest`: GeoJSON coordinates nodes list.
- GET `/api/alerts`: Active alerts list.
- GET `/api/geofences`: Defined restricted polygon coordinates.

REFERENCES

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- [3] Docker Inc., "Docker Compose File Format specification v3.8," 2023.
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